4.3 $22/7$ Is Bigger Than π

WORKSHEET

The question

STEP I 1997, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why does π come out of $\int_0^1 \frac{\text{polynomial}}{1+x^2}\,dx$? Which simple integral of this form gives π on its own?

STAGE 1 Polynomial division

Covers: the division by x^2+1

Ask yourself Expand $x^4(1-x)^4$ first.

Pause and try Divide by x^2+1, by long division or by comparing coefficients from the top. What is the remainder?

STAGE 2 The integral

Covers: the integral equal to $\frac{22}{7} - \pi$

Ask yourself Divide your identity by $1 + x^2$ and integrate term by term from 0 to 1. Which integral gives π?

STAGE 3 Bounding the integral

Covers: the bounds on π

Ask yourself The integrand is positive on $(0,1)$. What does that tell you straight away?

Pause and try How large and how small can $\frac{1}{1+x^2}$ be on $[0,1]$? Use that to bound the integral, and evaluate $\int_0^1 x^4(1-x)^4\,\mathrm{d}x$.

What's really going on

A proof with a history

Archimedes proved $\frac{223}{71} < \pi < \frac{22}{7}$ around 250 BC by computing the perimeters of 96-sided polygons inside and outside a circle (compare the polygons of Section 2.5). The integral proof is much more recent. It was published by D. P. Dalzell in 1944, and became famous when it was set as a problem in the 1968 Putnam competition for American undergraduates. Its charm is that it needs no numerical work at all: the inequality $\frac{22}{7} > \pi$ drops out of the fact that an integral of a positive function is positive.

Better approximations from the same idea

The method generalises. With $x^8(1-x)^8$ in place of $x^4(1-x)^4$, the same division gives

$$\int_0^1 \frac{x^8(1-x)^8}{1+x^2}\,\mathrm{d}x = 4\pi - \frac{188684}{15015},$$

which proves $\pi > \frac{47171}{15015} = 3.1415917\ldots$, correct to six significant figures. In 2009 Stephen Lucas found families of integrals of this kind with non-negative integrands that prove inequalities for other famous approximations, including $\frac{355}{113}$, which is correct to seven significant figures.

Approximations and irrationality

Why does π have such good fractional approximations as $\frac{22}{7}$ and $\frac{355}{113}$? Every irrational number has infinitely many fractions $\frac{p}{q}$ within $\frac{1}{q^2}$ of it, and π happens to have some exceptionally good ones early on, because its continued fraction $[3;7,15,1,292,\ldots]$ contains the large entry 292 (see Section 2.2). Integrals like the one in this question, made smaller and smaller, are the starting point for proofs that numbers *cannot* be approximated too well by fractions, and so are irrational: Ivan Niven's famous one-page proof that π is irrational (1947) uses integrals of $\frac{x^n(1-x)^n}{n!}\sin(\pi x)$, the trigonometric cousins of those in Section 2.9.

Try this next

1. **Check the numbers.** Compute $\frac{22}{7} - \pi$ to four significant figures and confirm that it lies between $\frac{1}{1260}$ and $\frac{1}{630}$.

2. **A Beta integral.** Use the formula $\int_0^1 x^n(1-x)^n\,\mathrm{d}x = \frac{(n!)^2}{(2n+1)!}$ (proved in Section 4.4) to check that $\int_0^1 x^4(1-x)^4\,\mathrm{d}x = \frac{1}{630}$.

3. **A lower approximation.** Show that

$$\int_0^1 \frac{x(1-x)^2}{1+x^2}\,\mathrm{d}x = \frac{\pi}{2} - \frac{3}{2} \quad\text{and}\quad \int_0^1 \frac{x^2(1-x)^4}{1+x^2}\,\mathrm{d}x = \pi - \frac{47}{15},$$

   and deduce that $\pi > 3$ and $\pi > \frac{47}{15}$. Why are these less accurate than $\frac{22}{7}$?

4. **Going further.** HARDER Verify the identity for $\int_0^1 \frac{x^8(1-x)^8}{1+x^2}\,\mathrm{d}x$ given above. (A computer algebra system will do the division; the point is to see that the remainder is 16, giving 4π.)

Further reading

- **Petr Beckmann, *A History of* π** (St Martin's Press). Archimedes' polygon method in detail, and much else.

- **Stephen Lucas, "Approximations to π derived from integrals with nonnegative integrands"**, *American Mathematical Monthly* 116 (2009). Short and readable.
- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). Includes Niven's proof that π is irrational.

The full guided solution is in Section 4.3:
https://beyondstep.keithmcnulty.org/ch04/twenty-two-sevenths.html